VoltGuard: A Microcontroller-Based Voltage and Overcurrent Protection System

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GitHub: https://github.com/Ahsan-Habib147/VoltGuard-A-Microcontroller-Based-Voltage-and-Overcurrent-Protection-System

Design and Implementation: I designed VoltGuard in Proteus using an Arduino Uno to protect a simulated load from abnormal voltage and overcurrent conditions. Two switches select between normal, sag, and surge voltage sources. The voltage-sensing circuit uses two 470 kΩ resistors and a 5.1 kΩ resistor as a divider, with a bias network to center the signal around 2.5 V. The firmware samples A0 for 40 ms, calculates the signal variance, and applies a scaling factor of 204.65 to estimate RMS voltage.

The current-sensing input is connected to A1. The firmware calculates the average absolute deviation from the 2.5 V reference and divides it by the configured sensitivity of 0.185 V/A. This provides a current-related estimate rather than true RMS current. The default protection limits are 280 V for overvoltage, 170 V for undervoltage, and 10 A for overcurrent, with a 120 ms trip delay. A 16×2 I²C LCD, four push buttons, EEPROM-based settings storage, and serial communication at 9600 baud provide local control and computer monitoring.

Results and Discussion: During the Proteus simulation, the normal source displayed approximately 229–230 V and ON. The surge source produced approximately 280.8 V and TRIP OV (fault code 1), while the sag source produced approximately 168–170 V and TRIP UV (fault code 2). During the overcurrent test, the displayed current-related value reached approximately 10.3 A, producing TRIP OC (fault code 3). When the voltage returned to the normal range, the relay reconnected after approximately 10 seconds with the configured recovery delay. A 5 V hysteresis margin helps prevent repeated switching near the voltage limits.

The overcurrent trip remains latched and requires a manual RESET even after the fault disappears, preventing automatic reconnection to a potentially overloaded load. The OV limit can be adjusted using the push buttons or the serial command SET OV, the updated setting can then be checked using GET. EEPROM storage is implemented, but persistence across a simulation restart has not been independently verified. The current input uses a potentiometer rather than a calibrated current sensor, so actual load-current accuracy has not been established. No physical isolation or surge testing was conducted.

Conclusion: VoltGuard combines analog measurement, configurable protection, relay control, an LCD, local settings, and computer communication in a simulated embedded system. The simulation demonstrates the intended fault-detection workflow, while calibrated sensing and further validation are needed before practical deployment. No mains-connected prototype was built.